


**Ministry of Higher Education**

**Soran University**

Part 3 - System Modeling and Design

# Second-hand Item Exchange System

**GROUP MEMBERS**

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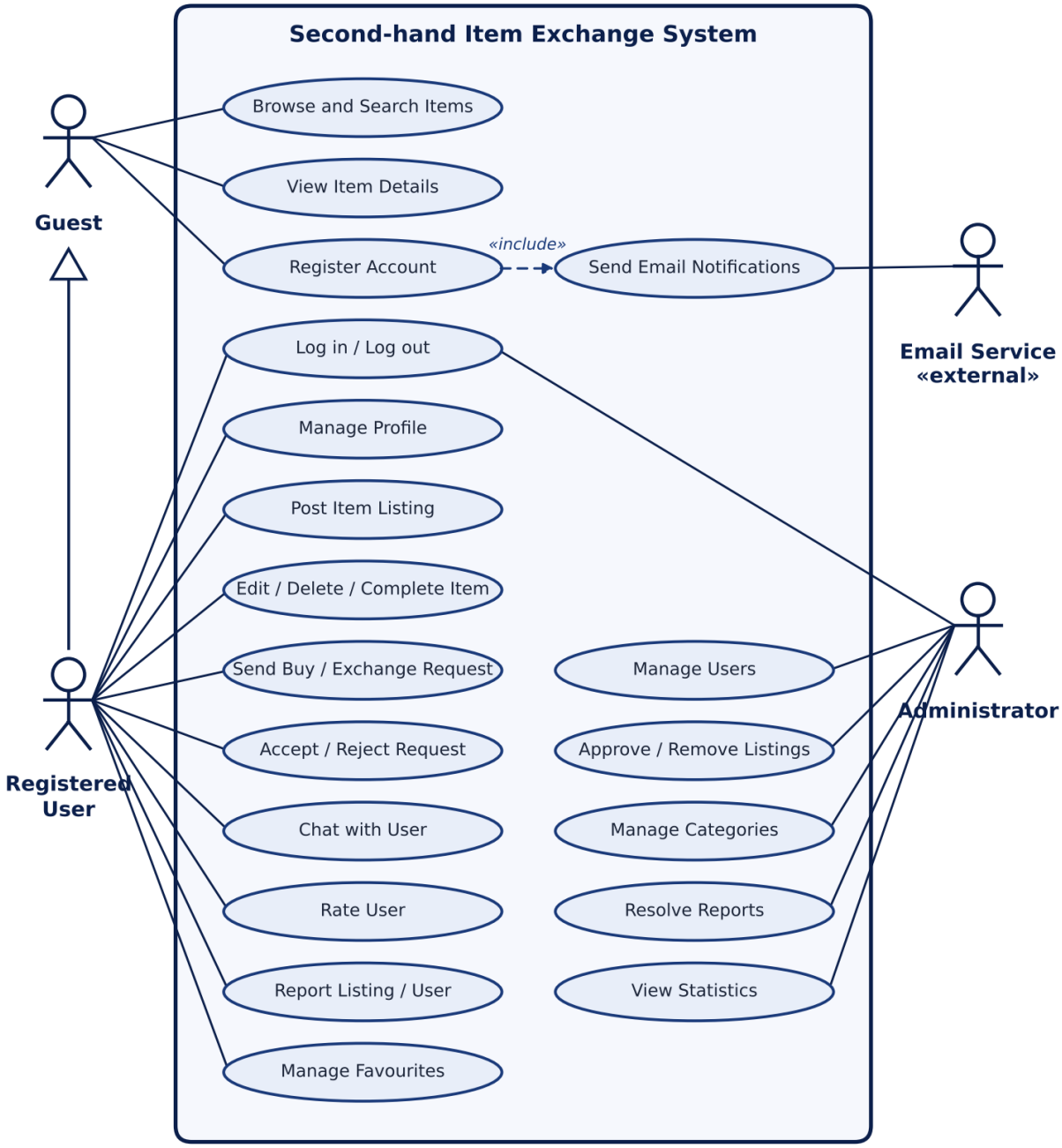
**SUPERVISOR**

Dr. Sirwan

## 1. Use Case Diagram

This diagram shows who uses the system and which functions each user can perform. The actors are the Guest, the Registered User, and the Administrator, plus the external Email Service. A Registered User inherits everything a Guest can do, while the Administrator manages users, listings, categories, and reports. It helps the group agree on the system functions before design and testing.

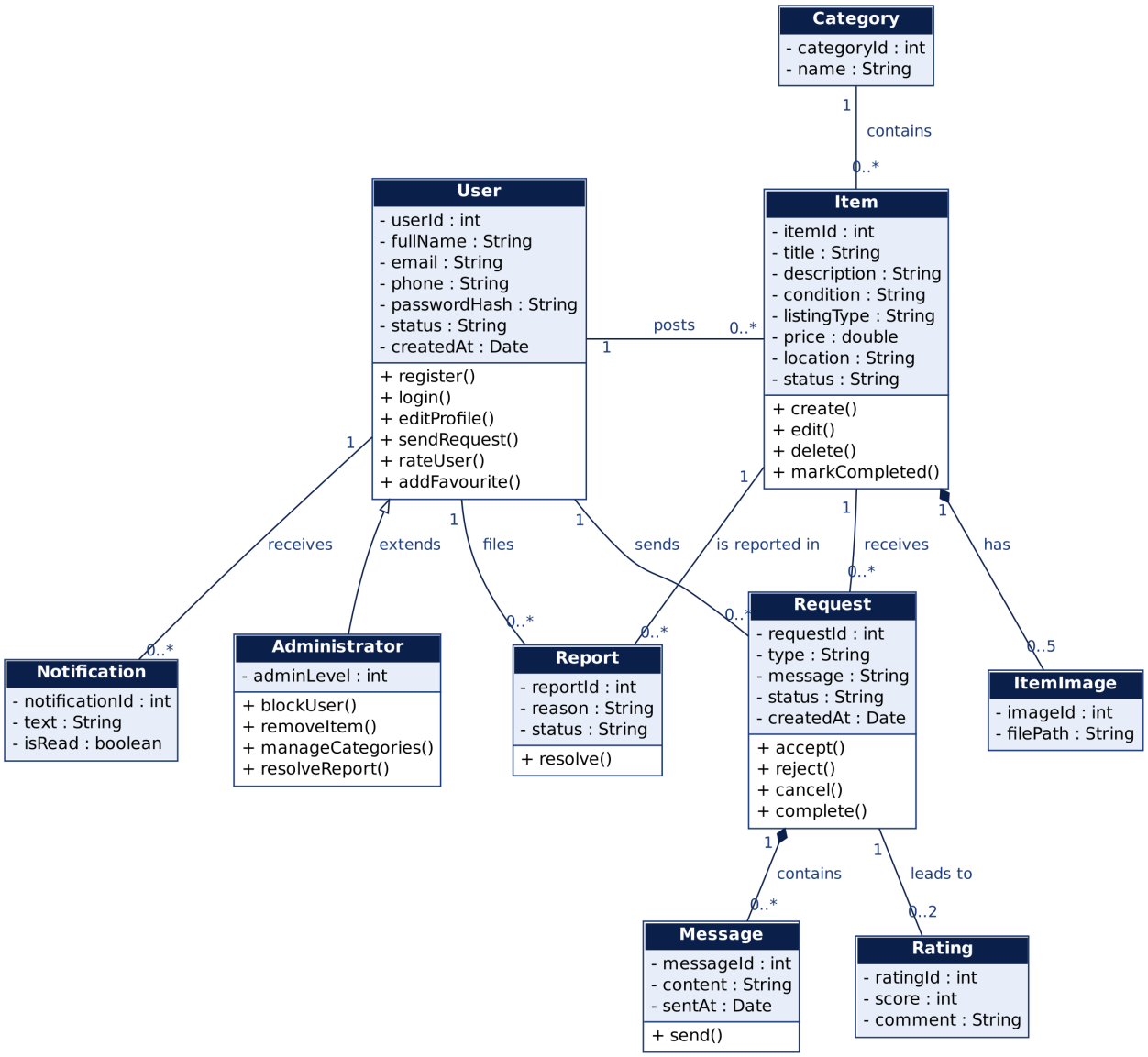
**Supports SRS requirements:** all functional requirements FR-01 to FR-27 (account, listings, search, requests and messaging, ratings and reports, administration), user requirements UR-01 to UR-12, and the three user roles in Section 3.



## 2. Class Diagram

This diagram shows the main classes of the system, their key attributes and methods, and how they are related. A User posts Items, sends Requests, writes Messages, and rates other users; an Administrator is a special kind of User. Each Item belongs to a Category and has up to five images, and each Request can lead to two Ratings. The diagram helps the group design the code structure and the database.

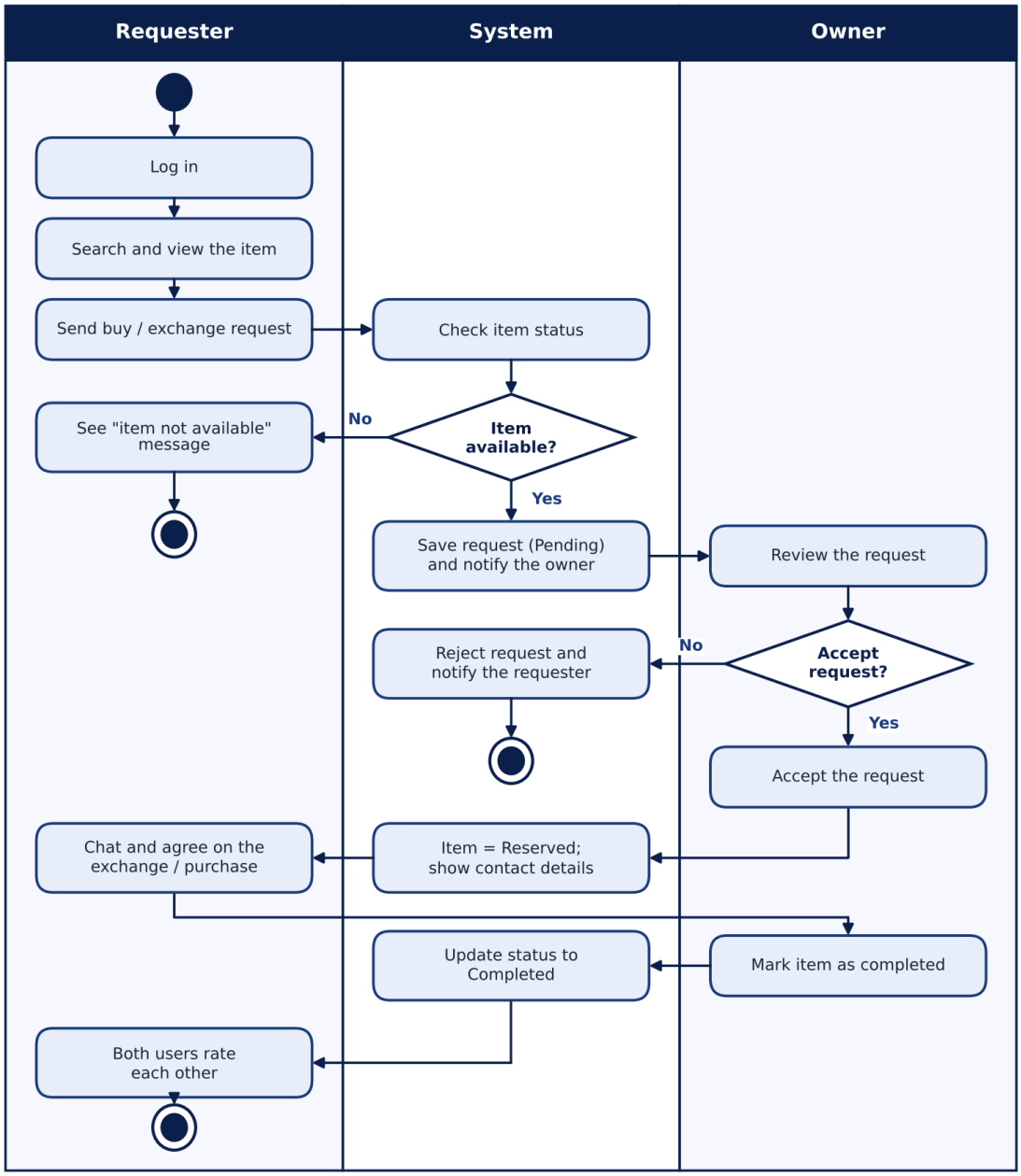
**Supports SRS requirements:** FR-01 to FR-05 (accounts), FR-06 to FR-09 (items), FR-15 (favourites), FR-16 to FR-20 (requests, messages, notifications), FR-21 and FR-22 (ratings, reports), FR-23 to FR-26 (administration), NFR-03 (password hash).



## 3. Activity Diagram - Buy / Exchange Request

This diagram models the workflow of sending and completing a buy or exchange request, shown in three swimlanes: Requester, System, and Owner. The main decisions are whether the item is still available and whether the owner accepts the request. After acceptance the item is reserved, contact details are shown, and both users rate each other when the exchange is completed. It helps the group understand the process and the status changes before implementing it.

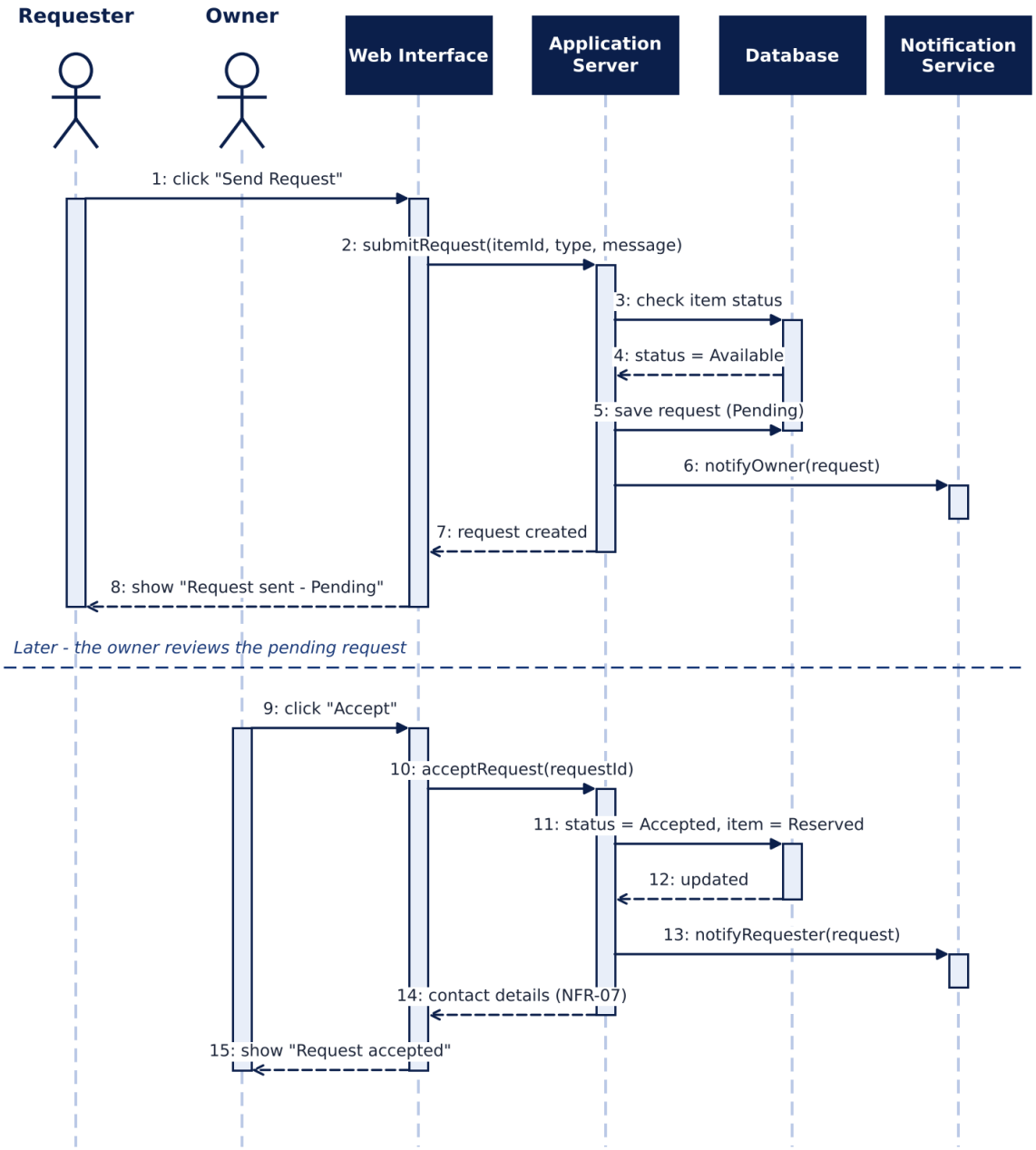
**Supports SRS requirements:** FR-03, FR-08, FR-09, FR-10, FR-11, FR-16 to FR-21; user requirements UR-06 to UR-09; NFR-07 (contact details only after acceptance).



## 4. Sequence Diagram - Send and Accept a Request

This diagram shows, over time, how the Requester, the Owner, and the system components interact when a buy or exchange request is made and accepted. The Web Interface passes the request to the Application Server, which checks the item, stores the request in the Database, and asks the Notification Service to inform the other user. The scenario is the same one described in the activity diagram and the SRS. It helps the group see which component is responsible for each step.

**Supports SRS requirements:** FR-09, FR-16, FR-17, FR-18, FR-20; user requirements UR-06 and UR-07; NFR-01 (response time), NFR-07 (contact details after acceptance).



## 5. Database / ER Diagram

This diagram shows the main database tables, their columns, and the relationships between them. Primary keys (PK) identify each row and foreign keys (FK) link the tables, for example items.owner_id points to users.user_id. The favorites table links users and items in a many-to-many relationship. The ER diagram matches the class diagram and is the base for creating the database in the implementation phase.

**Supports SRS requirements:** FR-01, FR-06, FR-10 to FR-12, FR-15 to FR-22, FR-25; NFR-03 (password_hash); Section 5 (data storage).

